

GOLMUD, China

WMO: 528180

Lat: 36.4207N

Lon: 94.9083E

Elev: 2809

StdP: 71.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-14.3	-29.4	0.3	-7.7	-28.0	0.4	-7.2	6.3	-4.4	5.5	-5.0	1.3	270	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	28.5	12.3	26.8	11.4	25.2	10.8	14.0	25.0	12.8	23.1	12.0	22.4	2.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
10.0	10.8	18.4	8.7	9.9	17.6	7.4	9.1	17.1	49.5	25.0	45.8	23.0	43.0	22.6	18.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.6	5.7	5.0	-18.8	31.9	1.2	1.3	-19.6	32.8	-20.4	33.6	-21.1	34.4	-21.9	35.4		
			-19.9	15.5	1.1	1.4	-20.7	16.5	-21.3	17.3	-22.0	18.1	-22.8	19.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-6.9	-2.7	2.4	8.3	12.9	16.7	19.2	18.5	13.8	6.9	-0.4	-6.1	(6)
(7)		DBStd	9.61	3.04	3.54	3.61	3.69	2.98	2.61	2.73	2.50	2.63	3.07	3.28	2.73	(7)
(8)		HDD10.0	2118	524	356	236	74	9	1	0	0	2	105	313	498	(8)
(9)		HDD18.3	4259	782	589	493	302	169	59	23	28	137	356	563	757	(9)
(10)		CDD10.0	998	0	0	1	23	99	202	284	264	117	8	0	0	(10)
(11)		CDD18.3	96	0	0	0	0	1	11	48	34	2	0	0	0	(11)
(12)		CDH23.3	838	0	0	0	5	18	106	413	274	22	1	0	0	(12)
(13)		CDH26.7	156	0	0	0	1	1	9	93	51	1	0	0	0	(13)
(14)	Wind	WSAvg	2.2	1.8	2.0	2.4	2.6	2.7	2.6	2.5	2.3	2.1	1.9	1.7	1.7	(14)
(15)	Precipitation	PrecAvg	41	1	0	1	1	5	8	11	8	4	1	1	1	(15)
(16)		PrecMax	102	3	3	8	12	23	27	44	34	14	14	4	5	(16)
(17)		PrecMin	11	0	0	0	0	0	0	1	0	0	0	0	0	(17)
(18)		PrecStd	18	1	1	2	2	6	6	9	8	5	2	1	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	11.6	18.2	23.4	25.5	28.0	31.3	30.2	25.9	20.6	12.4	5.7	(19)
(20)		MCWB	-2.6	0.5	3.9	6.7	8.4	11.3	14.0	13.1	11.2	6.3	1.3	-2.1	-2.1	(20)
(21)		2%	DB	3.0	8.7	14.1	20.3	23.2	25.9	29.1	28.0	23.3	17.4	10.1	3.1	(21)
(22)		MCWB	-4.2	-1.0	1.9	5.5	7.4	10.4	12.8	12.2	9.5	4.7	0.1	-3.8	-3.8	(22)
(23)		5%	DB	1.0	6.3	11.8	18.0	21.4	24.4	27.4	26.2	21.5	15.4	8.1	1.2	(23)
(24)		MCWB	-5.3	-2.3	0.8	4.4	6.7	9.6	11.9	11.4	8.6	3.8	-0.9	-4.9	-4.9	(24)
(25)		10%	DB	-1.0	4.1	9.8	15.9	19.8	23.0	25.6	24.6	19.8	13.4	6.2	-0.5	(25)
(26)		MCWB	-6.4	-3.4	-0.3	3.4	6.0	9.0	11.3	10.7	7.7	2.9	-1.8	-5.9	-5.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	0.6	4.1	7.2	9.2	13.3	15.9	15.3	13.4	7.1	1.5	-1.9	(27)
(28)		MCDWB	5.3	11.1	17.5	21.0	21.5	23.4	28.5	27.2	22.1	17.5	11.9	5.7	5.7	(28)
(29)		2%	WB	-4.0	-0.9	2.1	5.9	8.3	11.7	14.2	13.8	11.1	5.5	0.1	-3.8	(29)
(30)		MCDWB	2.6	8.0	13.1	19.2	20.3	22.0	25.9	24.1	20.0	15.5	9.5	2.6	2.6	(30)
(31)		5%	WB	-5.1	-2.1	1.0	4.7	7.4	10.8	13.1	12.7	9.9	4.3	-0.8	-4.8	(31)
(32)		MCDWB	0.8	6.0	11.4	17.1	19.2	21.1	24.0	22.6	19.2	14.2	7.7	0.9	0.9	(32)
(33)		10%	WB	-6.3	-3.3	-0.1	3.7	6.6	10.0	12.2	11.7	8.7	3.3	-1.8	-5.8	(33)
(34)		MCDWB	-1.1	3.9	9.4	15.1	18.1	20.4	23.0	21.9	17.9	12.8	5.9	-0.7	-0.7	(34)
(35)	Mean Daily Temperature Range		MDBR	11.6	12.4	12.7	12.5	11.6	10.6	11.0	11.1	11.5	12.5	12.3	11.5	(35)
(36)		5% DB	MCDDBR	13.9	15.1	15.9	15.6	14.1	13.0	13.9	13.8	13.9	14.9	14.1	13.1	(36)
(37)			MCWBR	9.1	8.8	8.4	7.3	6.3	5.5	5.5	5.8	6.0	7.0	7.9	8.4	(37)
(38)			MCDDBR	13.4	14.4	15.1	14.6	12.5	11.2	12.1	11.6	11.7	13.7	13.6	12.6	(38)
(39)		5% WB	MCWBR	8.8	8.5	8.0	6.9	5.6	4.6	4.9	5.0	5.3	6.5	7.6	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.229	0.271	0.336	0.395	0.408	0.406	0.397	0.384	0.350	0.296	0.248	0.226	(40)	
(41)		taud	2.464	2.333	2.091	1.928	1.971	2.070	2.173	2.198	2.263	2.345	2.438	2.480	(41)	
(42)		Ebn at Noon	994	973	924	884	876	875	880	886	901	929	957	977	(42)	
(43)		Edh at Noon	81	106	149	186	182	165	148	141	124	103	83	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.25	4.14	5.54	6.53	7.05	6.81	6.83	6.40	5.50	4.72	3.63	3.03	(44)	
(45)		RadStd	0.15	0.18	0.15	0.25	0.27	0.45	0.40	0.32	0.33	0.16	0.13	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.42	N/A	N/A	N/A	N/A	N/A	-113	-152	+54	N/A
(47)	Regional (1 neighbor)	+0.48	N/A	N/A	N/A	N/A	N/A	-127	-168	+54	N/A

Nomenclature: See separate page